

Insuring Against Extreme Heat: A Randomised Evaluation

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Climate change is causing extreme heat now

- ▶ In India in 2024:
 - 37 cities surpassed 45°C (113°F)
 - 500 out 741 districts hit 40°C at least once in April/May
 - 3 districts in Maharastra hit 40°C on 48 of 61 days.
- ▶ Extreme heat has negative physical, economic and emotional impacts (e.g. Burgess et al. 2017)
- ▶ Should social protection systems change to reflect new climate risks?

This paper

- ▶ RCT of parametric heat insurance: pays daily wage when temperature passes threshold (41.6°C)
 - Highly subsidised (pay INR300, get INR400 + heat payments)
 - Implemented by the Self-Employed Women's Association (SEWA) in India: 7 states 250,000 women
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 - Baseline data & participation from SEWA membership database
 - Outcomes measured through surveys in Sept-Nov 2025

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 - Baseline data & participation from SEWA membership database
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- ▶ Big picture: Can it exist without subsidies? If not, is it a good use of public / philanthropic funds?

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 - Take actions to reduce exposure/damages (e.g. use aircon / drink water / reduce work hours)
- ▶ Why might government intervene:
 - Missing insurance market → support (make obligatory, subsidise) or provide (social insurance)
 - Inefficiently low adaptation → information / subsidies / fix credit markets

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- ▶ Maybe this is more feasible? Why do we ever target particular risks?
- ▶ Key climate adaptation question: do something different or make existing systems work better.

The Self-Employed Women's Association has 3.2 million members



Vendors and Hawkers



Home-based Workers



Labour & Service Providers



Producers (e.g. farmers)

Qualitative impacts of extreme heat on SEWA members

Consistent with empirical literature

▶ Heat-related health impacts

- headaches, dizziness, fainting, vomiting, fungal infections, muscle cramps, rapid pulse, fever, UTIs
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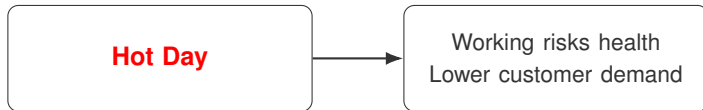
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 - Severe health consequences from working during hottest hours of the day
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- ▶ Loss of income → financial and emotional distress
 - High interest informal loans, sale of productive assets
 - Mental stress of financial challenges and increased care responsibilities

Responses to a hot day

Hot Day

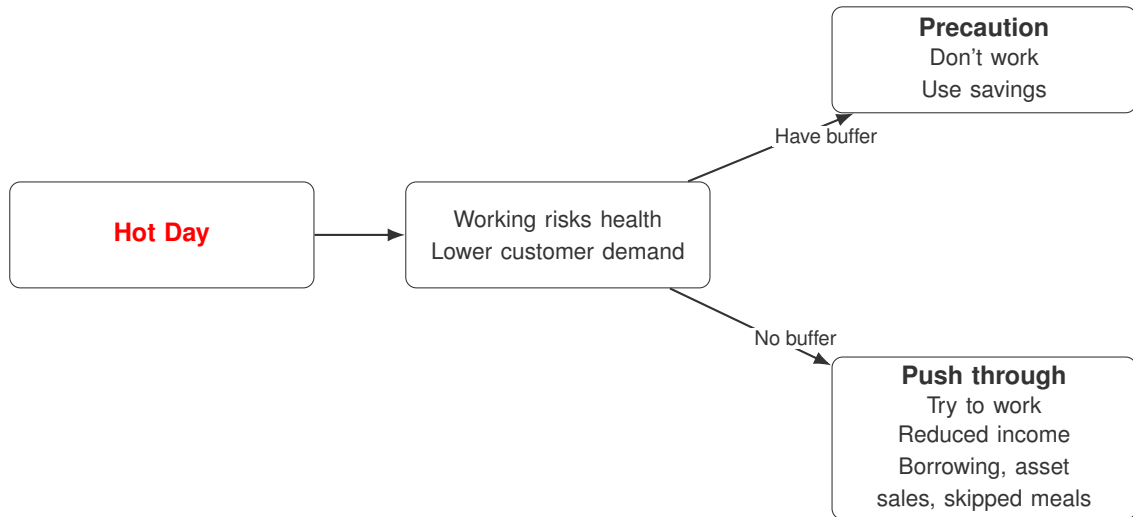
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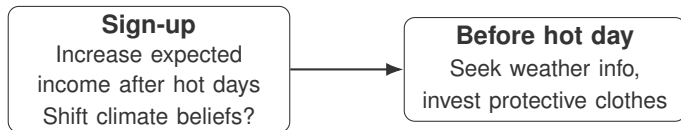
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- ▶ A signal that a threshold exists beyond which conditions are especially hazardous
- ▶ A message of support - conveying that “someone is looking out for me” during climate stress

A Theory of Change

Sign-up

Increase expected
income after hot days
Shift climate beliefs?

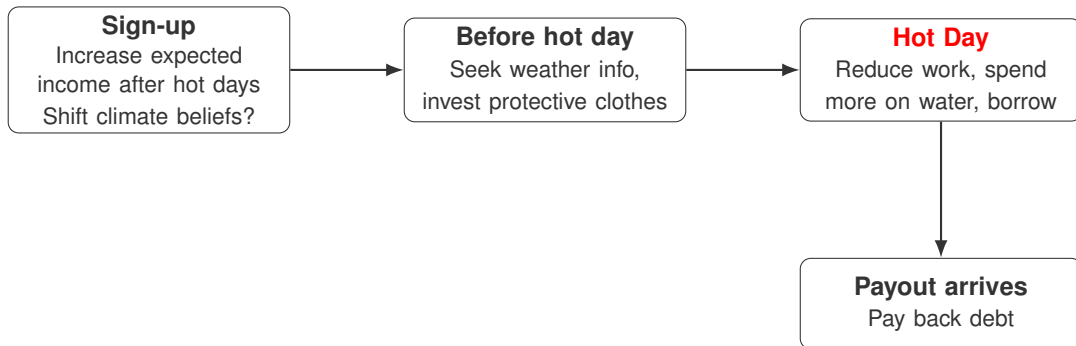
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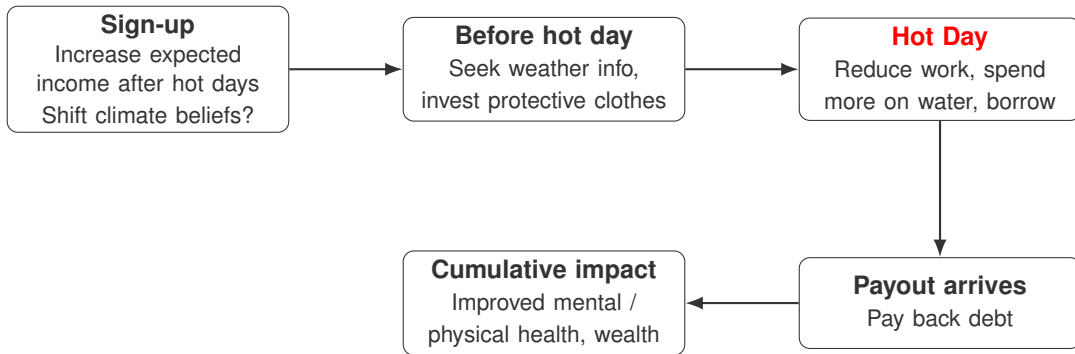
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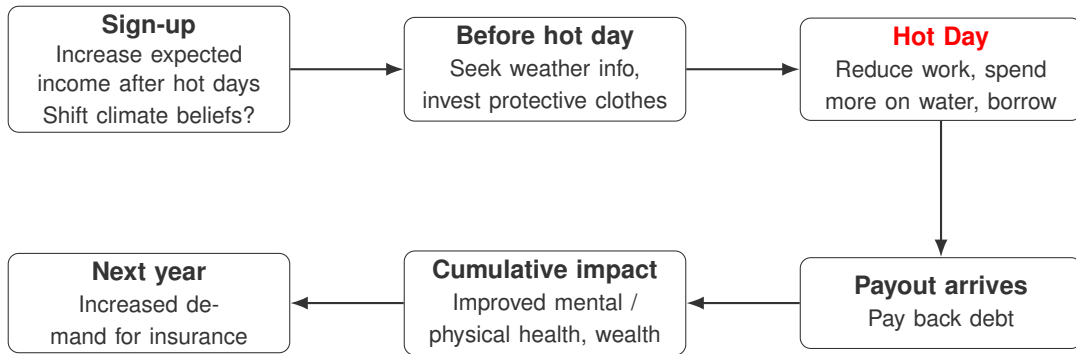
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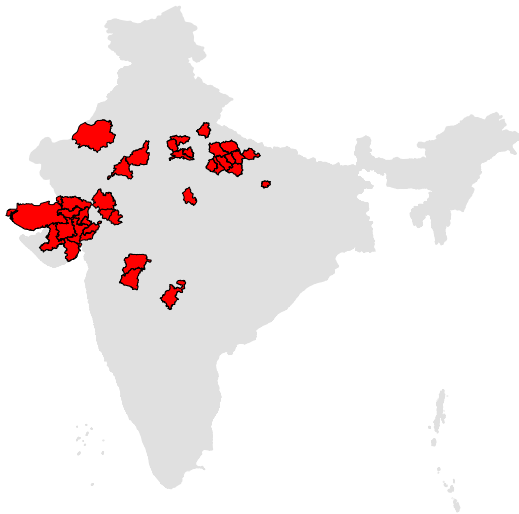
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 - Villages are randomly assigned to treatment and added to a “priority list.”
 - Leaders receive incentives to recruit only in priority (treatment) villages.

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- ▶ To generate exogenous variation in take-up:
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 - Leaders receive incentives to recruit only in priority (treatment) villages.
- ▶ Use village-level treatment assignment as an IV for individual participation.

Study covers 5,631 villages in Gujarat, UP, Rajasthan, and Maharashtra



Key outcomes

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- ▶ **Use of financial coping strategies** e.g. skipping meals, borrowing, or selling assets.

Secondary outcomes

- ▶ Heat-related health problems
- ▶ Mental health
- ▶ Healthcare use (expenditure and visits)
- ▶ Assets and liabilities
- ▶ Consumption and income
- ▶ Women's empowerment
- ▶ Climate beliefs and willingness to pay for insurance
- ▶ Understanding of insurance mechanisms
- ▶ Spillovers on household members (e.g., labour supply, education)

Heterogeneity analysis (examples)

► **Occupation:**

- Differences in workplace, conditions and flexibility impact (i) heat exposure (ii) avoidance potential.
- Test for heterogeneity by occupational group, 'heat exposure' and market power.

► **Access to credit and savings:**

- Insurance may benefit those lacking financial buffers the most.
- Test for heterogeneity by baseline income and local credit market access.

► **Prior exposure to heat events:**

- Past heatwave exposure may shift perceptions and responses.
- Construct weather histories from secondary data.

Number of payments RDD

- ▶ Leverage payout threshold to implement regression discontinuity design (RDD).
- ▶ Compare “just eligible” vs. “just ineligible” by simulating eligibility under slightly adjusted thresholds.
 - **Control group:** villages that would have received an additional payout under a higher threshold.
 - **Treatment group:** villages that would have received one fewer payouts under lower threshold.